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FULL PAPER

Flexible needle posture control stratagem for ultrasound-based puncture manipulator system

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ABSTRACT

This paper introduces a flexible puncture method that can improve the efficiency and safety of robot-assisted central venous catheterization (CVC) surgery systems. Current CVC surgery-assisted manipulators and systems are mainly focused methods for puncture needle insertion into blood vessels. There is no single systematic stratagem for safe high-precision control of the posture of the needle in a blood vessel after needle insertion. However, this type of control stratagem is both necessary and urgently required in surgery to avoid piercing the central vein and to provide a suitable posture for guide wire insertion. Therefore, we propose the remote centre of motion control method to control the needle's posture precisely and safely. Verification experiments were performed in both a simulator and a puncture manipulator. The results indicate that the needle tip trajectory in the simulation is completely accurate and it also agrees well with the theoretical value. The positional and angular errors of the needle tip are smaller than 0.8 mm and 1.8°, respectively, in the real puncture manipulator experiment; these errors are sufficiently small for surgeries. The proposed design is practical and can be used in real puncture manipulators and systems.

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Puncture assistance system; remote centre of motion; ultrasound image navigation; medical robot; image processing

1. Introduction

With the continuing rapid development of robotic technology, increasing numbers of robot systems are being introduced into the medical field [1,2]. Central venous catheterization (CVC) is a technique that involves venous puncture and catheter insertion. CVC is an indispensable means of treatment for critically ill patients. Needle insertion is usually guided using 2D medical ultrasound (US) images. There are two important steps in CVC surgery, which are puncturing a blood vessel using a needle and then inserting a guide wire and catheter into the blood vessel. In traditional CVC surgery, the puncture needle is controlled by hand by a doctor. After the needle enters the blood vessel, the needle tip position and the needle's posture are adjusted to provide a suitable guided wire insertion angle [3,4]. Because of the shaking of the doctor's hands, tiredness after long operations and the varied levels of experience of the doctors, the resulting precision and puncture stability results could be very different.

Puncture manipulator systems have been introduced into some robotic CVC surgery systems to achieve better puncture accuracy and higher surgical safety. In current robot-assisted CVC surgical systems, researchers have mainly focused on the way to insert a puncture needle into a blood vessel using a needle. However, because of the robot's placement and individual differences in patients, the puncture angle of the needle is not always the best guide wire insertion angle. Because the central vein is close to arteries, it is dangerous for the central vein to be pierced. To insert the guide wire smoothly and successfully, the needle tip should be kept within the vein using a suitable posture. A CVC-assisted robot should thus have a function that allows it to adjust the posture of the puncture needle after it enters the blood vessel, similar to the doctor's flexible hands. To solve this problem, we propose a solution that offers good security and wide applicability called the remote centre of motion (RCM) control method for the needle tip during the adjustment period. Theoretically, when RCM control is applied, the control system will keep the needle tip at the same position in the central vein and ensure that the puncture needle is at a suitable angle to meet the requirements for guide wire insertion.

After the needle is inserted into the blood vessel, it is obvious that needle tip posture adjustment will also be affected by the resistance of the human tissues. Given that

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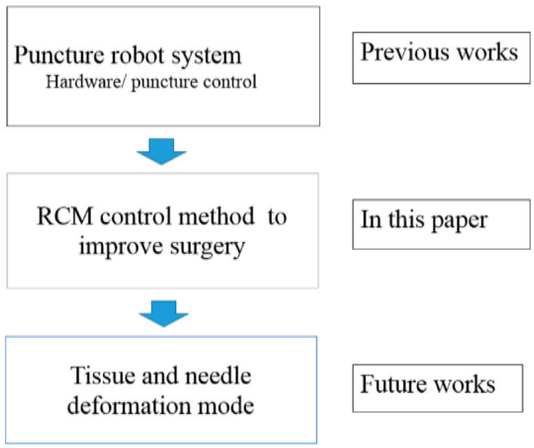


Figure 1. Research route.

the tissue structures from the skin to the blood vessels are highly complex, the deformation and resistance of each part of the tissues is different and thus cannot be simplified into a uniform linear model. Therefore, to apply RCM control in CVC surgeries, at least two tasks should be performed:

(1) Design a control algorithm to ensure that the puncture robot/system can realize RCM in a low resistance and low deformation environment, e.g. air or water. The aim of this step is to establish kinematic models and algorithms for puncture manipulator systems. This is the most basic and the most important step.
(2) Build a deformation model of human tissue from the skin to the blood vessels and then add deformation compensation to the algorithm in the first part of the work described above to ensure that the system will work properly during real puncture surgery. Theoretically, without a good deformation compensation method, precise control of the needle's posture is impossible.

Based on this analysis, our research route is illustrated in Figure 1. This paper will focus on the first step toward the design of the RCM control algorithm in a no-deformation test environment. To verify the proposed control stratagem, a puncture manipulator system [5] that was developed previously by our group was used in this paper as a hardware test platform. The rest of this paper is structured as follows. Related works are introduced in Section 2. The puncture manipulator system and the RCM control algorithm used for this manipulator will be described in detail in Section 3. Section 4 presents the experiments that were conducted to verify the correctness and effectiveness of the proposed design. In the final section, we draw conclusions from this paper and introduce works for the future.

2. Related works

Robot technology has been widely used in both research and medical care [6,7] and robot systems for assisting with puncture surgery have been developed and reported. Taillant et al. [8] designed a type of puncture robot architecture to perform interventional computed tomography (CT)/magnetic resonance (MR) procedures and punctures in particular, where the movements and positioning of the robot can be checked from CT or MR images. Bricault et al. [9] constructed the light puncture robot (LPR) with a compact body-supported architecture to follow the patient's body surface movements. The LPR is compatible with CT, open MR imaging (MRI) and closed MRI. Subsequently, a segmentation and registration algorithm and a control loop were added to localize the robot and verify its movements and positioning in their later work [10]. In addition to specially designed robots for puncture surgery, some researchers also tried to use universal robots with new control methods to allow these robots to be used in puncture surgery. Jiang et al. [11] designed a high-precision positioning method based on a particle swarm optimization (PSO)-optimized back-propagation (BP) neural network. This method was used in a universal robot UR3 manipulator and the experimental results indicated that the angle error of the end tool was less than 0.1°, while the position error was less than 0.1 mm.

To perform puncture surgery safely, surgeons need to be able to monitor and control the posture of the needle tip. Medical imaging is an efficient, convenient and low-cost way to realize this ability for most hospitals and research organizations. Using medical images, the relationship between the needle and the human body structure can be obtained easily. To date, medical imaging techniques including CT [12,13] images, MRI [14,15], US [16,17] and 3D US [18] have been used to guide puncture needles during surgery. These images provide operators with the ability to monitor and adjust needle postures during puncture surgery in real time.

For precise control of the position and trajectory of the needle tip during puncture processes, several important parameters have been investigated by different researchers. Kataoka et al. [19,20] studied the relationship between the needle deflection force and the needle penetration depth, then built a force-deflection model. Their experimental results indicated that the moment and the rotational force acting on the needle should also be considered in their force-deflection model. Abolhassani et al. [21] investigated the needle trajectory during puncture

processes in soft tissue and found that the infinitesimal force per displacement is a useful parameter for online trajectory generation. Seifabadi et al. [22] designed a teleoperated needle steering system for use in prostate intervention surgery. They proposed a robot-assisted bevel-tip needle steering method to compensate for target displacement during the puncture process. Webster et al. [23] also developed a needle steering mechanism to control the needle trajectory and avoid obstacles in the needle's path when puncturing soft tissue. Their experimental results showed that the bevel tip angle affects the needle deformation strongly. Kobayashi et al. [24] designed a needle tip position control stratagem for use in liver surgery. They combined a force sensor and a liver model to control the needle tip position, and test results indicated that this method was the most effective among their proposed approaches. Researchers have developed a variety of control methods based on their hardware; however, no single general control method was proposed and implemented for the needle posture adjustment process.

Similarly, the RCM technique is not new in medical applications. RCM has been used in other medical fields, including minimally invasive surgery [25–27]. One of the most famous surgical robots in the world, the *da Vinci* [28], uses the RCM control stratagem to minimize surgical trauma. The RCM control method has also been used in puncture robot systems [29] to increase the user-friendliness of robot operation. Basically, current RCM techniques are mainly designed for use with rigid medical tools, where the deformation of the surgical instruments during surgery can be ignored. However, the deformation of the needle and the tissue is large enough to affect puncture surgery results, so the deformation of both the tissue and the needle should be processed accurately. As this paper describes only part of the work in our research, the soft tissue deformation model will be introduced in another work and we will not focus on it in this paper.

3. Puncture manipulator system

To aid in the design and verification processes, a puncture manipulator system was used in the experiments. The system details have been reported in [5] and we only provide the necessary introduction in this paper to help the reader to understand it easily. This puncture manipulator system includes a control box, a control interface, an ultrasound device and the manipulator. The electrical equipment for manipulator control is collected in the control box. Operators input control commands via the user interface. The ultrasound probe is fixed in the manipulator to obtain ultrasound images of the tissue to be observed. The design and the RCM control algorithm are closely related to the manipulator structure and its main parts and structure are described in the following sections.

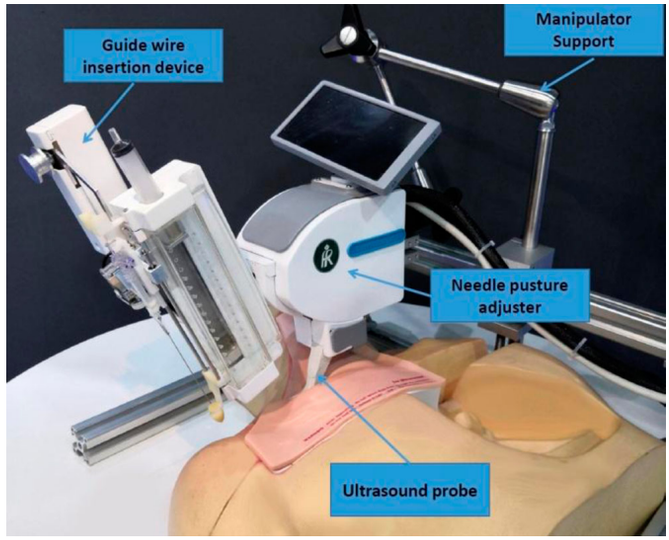


Figure 2. Puncture manipulator.

3.1. Puncture manipulator

The role of the puncture manipulator is to hold the puncture needle and insert it at a target position acquired by navigation based on ultrasound images. The manipulator consists of three parts: a needle posture adjuster, a guide wire insertion device and an ultrasound probe, as shown in Figure 2.

(1) Needle posture adjuster: This is a three-degrees-of-freedom (3-DOF) device that includes two translational DOFs and one rotational DOF; a schematic of this device is shown in Figure 3. The joints are driven using a micro servo motor in combination with a synchronous pulley and a precision ball screw. The rotational DOF gives a puncture angle range from 15° to 85°, while the translational DOF allows the needle to move within the entire sound beam area of the ultrasound probe. By controlling these joints, the device can achieve functions that include adjustment of the needle posture and needle insertion or extraction.
(2) Guide wire insertion device: This device has guide rollers driven by a micro servo motor. The insertion and extraction of the guide wire can be implemented using the friction generated by the guide rollers.
(3) Ultrasound probe: This probe is used to acquire ultrasound images of the target area to provide real-time visual information. Because the relative relationship between the needle posture adjuster and the ultrasound probe is known, the acquired ultrasound images can be used to guide puncture surgery.

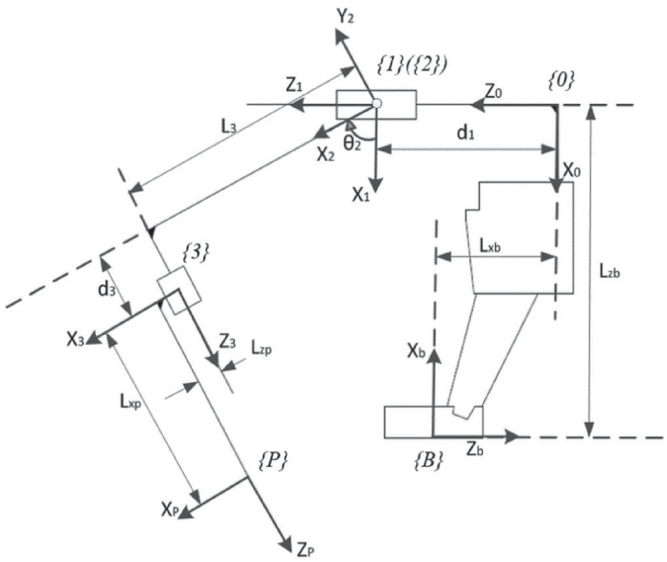


Figure 3. Schematic of puncture manipulator.

Table 1. D-H parameters of the manipulator.

Joint i	Link torsion angle $\alpha_{i-1}/(^\circ)$	Link length a_{i-1}	Link distance d_i	Joint angle $\theta_i(^\circ)$
0	180	L_{zb}	$-L_{xb}$	180
1	0	0	d_1	0
2	90	0	0	θ_2
3	90	L_3	d_3	0

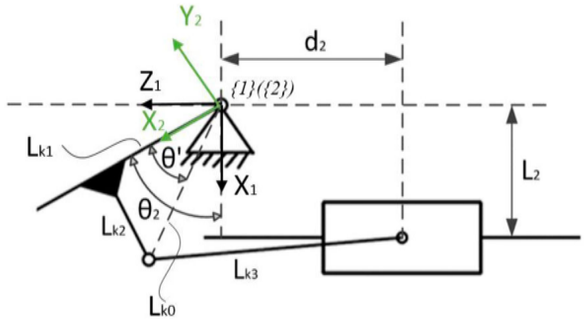


Figure 4. Rotation joint of puncture manipulator.

3.2. Kinematic analysis of puncture manipulator

The puncture robot manipulator can be simplified into the form of a 3-DOF series mechanism. In total, it has three movable joints, comprising two translational joints (joints 1 and 3) and one rotating joint (joint 2). The Denavit-Hartenberg (D-H) coordinate systems for this mechanism were established as shown in Figure 3 and the D-H parameters are listed in Table 1. The robot base coordinate system {***B***} is fixed using the ultrasound probe and the coordinate system origin is located at the centre point of the probe bottom. The puncture needle coordinate system {***P***} is fixed on the needle, where the coordinate system's origin is on the needle tip, and thus we can monitor this origin to obtain the needle's posture information. The other coordinate systems, {***0***}, {***1***}, {***2***} and {***3***}, are fixed in the corresponding joints. The existence of coordinate system {***0***} is for convenience of calculation and the joint 0 that is attached to this coordinate system is fixed, i.e. there is no rotational or translational motion of this joint. The coordinate origins of {***1***} and {***2***} are at the same position, while the coordinate origin of {***2***} will move with joint 1 (translational movement). The positive direction of each axis in the different coordinate systems is also marked in Figure 3. All these coordinate systems obey the right-hand rule and the axes that do not appear in these coordinate systems in Figure 3 can also be obtained using this rule. In the following parts of this paper, unless there are specific instructions, the coordinates mentioned are based on reference coordinate system {***B***}.

To reduce the size and weight of the manipulator, rotation joint 2 was designed to be a crank-slider mechanism; a schematic of this joint is shown in Figure 4.

The transformation matrix between the different coordinate systems can be obtained based on the robot structure and the D-H parameters.

The transformation matrix from {***0***} to {***B***} is as follows, where L_{zb} and L_{xb} refer to the distances between {***0***} and {***B***} in the X_b and Z_b directions, respectively, as shown in Figure 3.

$$\boldsymbol{T}_{\boldsymbol{B},0} = \begin{bmatrix} -1 & 0 & 0 & L_{zb} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & L_{xb} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1}$$

The transformation matrix from {***1***} to {***0***} is shown in Equation (2) and d_1 is the translational motion distance of joint 1 in the Z_1 direction.

$$\boldsymbol{T}_{0,1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

The transformation matrix from {***2***} to {***1***} is:

$$\boldsymbol{T}_{1,2} = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \sin\theta_2 & \cos\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

In addition, the transformation matrix from {***3***} to {***2***} is shown in Equation (4), where L_3 is the distance between coordinate systems {***2***} and {***3***} in the X_2 direction, and d_3 is the distance moved by joint 3 in the Z_3

direction.

$$\boldsymbol{T}_{2,3} = \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 0 & -1 & -d_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

The transformation matrix between {$\boldsymbol{P}$} and {$\boldsymbol{3}$} is shown in Equation (5). L_{zp} and L_{xp} are the needle tip offsets in the X_3 and Z_3 directions, respectively.

$$\boldsymbol{T}_{3,\boldsymbol{P}} = \begin{bmatrix} 1 & 0 & 0 & L_{\mathrm{zp}} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & L_{\mathrm{xp}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5}$$

Therefore, the transformation matrix from {$\boldsymbol{P}$} to {$\boldsymbol{B}$} can be obtained as:

$$\begin{aligned} \boldsymbol{T}_{\boldsymbol{B},\boldsymbol{P}} &= \boldsymbol{T}_{\boldsymbol{B},0}\boldsymbol{T}_{0,1}\boldsymbol{T}_{1,2}\boldsymbol{T}_{2,3}\boldsymbol{T}_{3,\boldsymbol{P}} \\ &= \begin{bmatrix} -\cos\theta_2 & 0 & -\sin\theta_2 & x_b \\ 0 & -1 & 0 & 0 \\ -\sin\theta_2 & 0 & \cos\theta_2 & z_b \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned} \tag{6}$$

The exact expressions for x_b and z_b in Equation (6) are given in Equations (7) and (8), respectively, in which x_b and z_b refer to the coordinates of the needle tip in {$\boldsymbol{B}$}:

$$x_b = -(L_3 + L_{zp})\cos\theta_2 - (d_3 + L_{xp})\sin\theta_2 + L_{zb} \tag{7}$$

$$z_b = -(L_3 + L_{zp})\sin\theta_2 + (d_3 + L_{xp})\cos\theta_2 + L_{xb} - d_1 \tag{8}$$

We would like to emphasize here that the rotation joint (joint 2) in this system is realized using a crank-slider mechanism. The stroke distance of the slider is d_2 and the corresponding rotational angle of joint 2 in the crank-slider mechanism is θ_2. From the structural relationship shown in Figure 4, Equation (9) can be obtained as:

$$\begin{aligned} {L_{k3}}^2 = {} & [L_{k0}\cos(\theta_2 - \theta') - L_2]^2 \\ & + [L_{k0}\sin(\theta_2 - \theta') + d_2]^2 \end{aligned} \tag{9}$$

Using Equations (7–9), the relationship between the displacements (d_1, d_2 and d_3) of the three motors and the needle tip position can be obtained, as shown in Equations (10–12). Therefore, the needle tip trajectory can be designed and by controlling the three displacements, the needle tip trajectory and the needle posture can both be controlled.

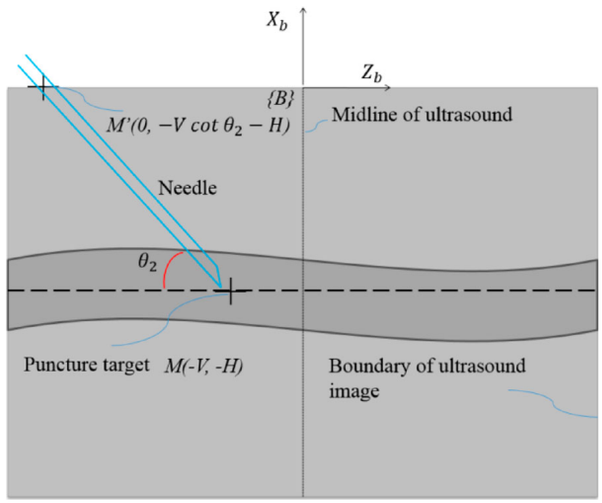


Figure 5. Puncture target point.

$$d_1 = F_1(z_b, x_b, \theta_2) = \frac{-x_b + L_{zb} - (L_3 + L_{zp})cos\theta_2}{\sin\theta_2} cos\theta_2 - (L_3 + L_{zp})\sin\theta_2 + L_{xb} - z_b \tag{10}$$

$$d_2 = F_2(\theta_2) = -L_{k0}\sin(\theta_2 - \theta') + \sqrt{{L_{k3}}^2 - [L_{k0}\cos(\theta_2 - \theta') - L_2]^2} \tag{11}$$

$$d_3 = F_3(x_b, \theta_2) = \frac{-x_b + L_{zb} - (L_3 + L_{zp})\cos\theta_2}{\sin\theta_2} - L_{xp} \tag{12}$$

3.3. Control of puncture trajectory

In the ultrasound-guided puncture system, the puncture needle tip can be observed easily using the ultrasound image. The puncture target is the point that the puncture needle tip must reach. Based on the experience of the doctor and the blood vessel detected in the ultrasound images, the doctor can specify a suitable puncture target M and puncture angle θ_{2f} via the control interface. Because the real size of the ultrasound image is known, after the puncture target is specified, the coordinates of point M *(–V, –H)* in {$\boldsymbol{B}$} can be obtained automatically using a simple image processing program, as shown in Figure 5. What we would like to clarify here is that the order of the coordinates of M is (x_b, z_b) in coordinate system {$\boldsymbol{B}$}. Therefore, V is the distance between M and the Z_b axis, while H is the distance between M and the X_b axis. To ensure that the needle tip can reach the specified position with the preset puncture angle, there are three steps that must be controlled precisely.

(1) *Adjust the manipulator to the puncture start position.* After the puncture target and the angle are specified, the puncture manipulator system will control the needle to

move from its initialization position ($x_b = x_{b0}$, $z_b = z_{b0}$, $\theta_2 = 90^\circ$) to the puncture start position ($x_b = 0$, $z_b = -V\cot\theta_2 - H$, $\theta_2 = \theta_{2f}$). The puncture needle tip will touch the skin when this step is completed, and the needle tip position will be $M'(0, -V\cot\theta_2 - H)$), while the angle of joint 2 (θ_2) is θ_{2p} (the ideal puncture angle), as shown in Figure 5. The process from the initialization position to the puncture start position can be described using three joint moments and the displacements of these three joints during this period can be expressed using the following equations:

$$\Delta d_1 = F_1(-V\cot\theta_{2p} - H, 0, \theta_{2p}) - F_1(z_{b0}, x_{b0}, 90^\circ) \tag{13}$$

$$\Delta d_2 = F_2(\theta_{2p}) - F_2(90^\circ) \tag{14}$$

$$\Delta d_3 = F_3(0, \theta_{2p}) - F_3(x_{b0}, 90^\circ) \tag{15}$$

(2) *Insert the needle into the blood vessel.*
If the manipulator is in the puncture start position, the puncture process can commence. The puncture needle will pierce the patient's skin at a constant angle θ_{2p} until the needle point reaches the specified puncture target M, which has coordinates of $(-V,-H)$. This means that the needle tip will move from M' to M, and Figure 5 shows the state of the needle when the needle tip arrives at the puncture target. During this period, joints 1 and 2 will maintain unchanged positions, and joint 3 will move forward to insert the needle with a specified speed. The joint displacements during this period can be described using the following equations:

$$\Delta d_1 = 0 \tag{16}$$

$$\Delta d_2 = 0 \tag{17}$$

$$\Delta d_3 = F_3(-V, \theta_{2p}) - F_3(0, \theta_{2p}) \tag{18}$$

(3) *Remote centre of motion method to adjust needle posture in blood vessel.*
To insert the guide wire smoothly, it is necessary to adjust both the position and the angle of the needle point in the blood vessel to obtain a suitable guide wire insertion angle. During the puncture angle adjustment process, the needle tip should remain in the vessel up to a specific length [30]. Otherwise, the needle may be removed from or pierce the blood vessel, which would cause puncture failure or even medical accident. This situation must be avoided in puncture surgery.

To reduce relative movement between the needle and the blood vessel, the RCM control method is introduced and applied. The RCM method is a control method in which the rotation centre (with no physical revolute joint contained therein) is located on the outer side of a real robot. The RCM method is advantageous in that it minimizes the risk of collision between the robot and the patient's body while simultaneously providing a sufficient range of movement to operate the instruments. This technology has thus been widely used in minimally invasive surgery.

During the needle angle adjustment process, we hope that the needle tip will remain motionless in the blood vessel, so we set the needle tip to be the RCM point. The purpose of the RCM method is to adjust the angle between the needle and the blood vessel (usually reducing this angle). We assume that the angle adjustment velocity of joint 2 is ω; therefore, after a specified period of time (t), the angle of joint 2 will be ($\omega t + \theta_{2p}$) and the needle tip position will be at $(-V,-H)$. The synchronous adjustment speed for the three joints during RCM movement can thus be expressed as follows:

$$v_1(t) = \frac{d}{dt}F_1[-H, -V, \omega t + \theta_{2p}] \tag{19}$$

$$v_2(t) = \frac{d}{dt}F_2(\omega t + \theta_{2p}) \tag{20}$$

$$v_3(t) = \frac{d}{dt}F_3[-V, \omega t + \theta_{2p}] \tag{21}$$

Based on the equations presented in Sections 3.3 and 3.4, we can control the manipulator system to finish the puncture process and determine the RCM moment. The same solution can be used for other puncture systems with different structures and would only require some parameters to be revised.

4. Experiments and analysis

The puncture trajectory and RCM control strategies were discussed in detail in the previous section. To verify the precision and the control accuracy of this design, two test experiments were performed. We must emphasize again here that this paper focuses on the RCM control method alone and we assume there is no deformation of the needle or the tissue during the puncture process. The following experiments were designed based on this assumption.

4.1. Experimental settings

The first experiment simulated the puncture and RCM processes in ADAMS software [31], while the second experiment involved testing the control system using the real puncture manipulator system.

(1) *Simulation in ADAMS*: ADAMS is a useful tool for dynamics and kinematics analyses. The software helps engineers to study the dynamics of moving parts and see how loads and forces are distributed throughout mechanical systems.

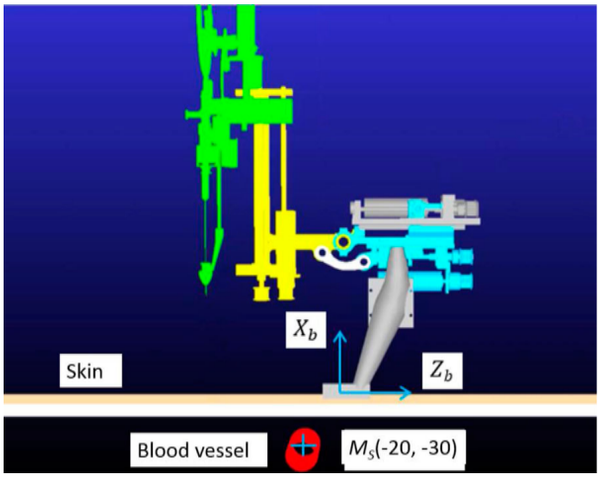


Figure 6. Experimental environment in ADAMS software.

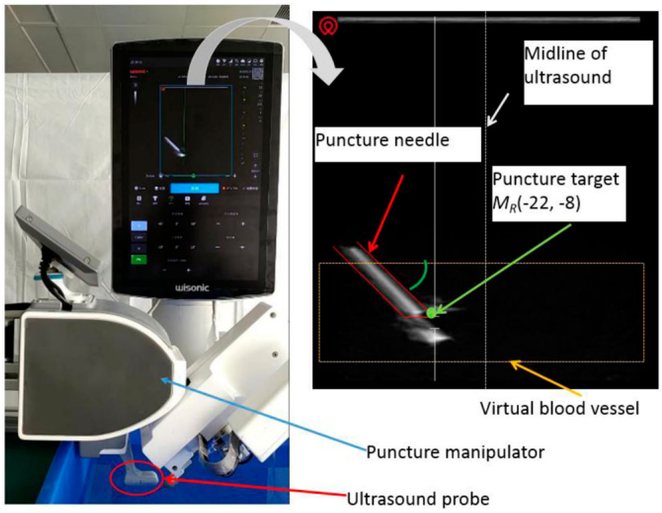


Figure 7. Test platform and ultrasound image.

Experimental environment: The experimental environment is shown in Figure 6. To simulate the motion of the puncture manipulator, the 3D model of the device must be input into ADAMS. The relevant motion constraints given in Equations (13–21) were then added for the puncture and RCM processes. A virtual cylindrical blood vessel was built in the model space, where the centre of this blood vessel in the *XBZ* plane of coordinate system {***B***} was M_S $(-20, -30)$ (units: mm) and this point was set as the puncture target.

Parameter setting: In the simulations, we do not need to worry about the manipulator breaking down, so we can test the manipulator system at maximum strength. We selected a large puncture angle $\theta_{2p}(60^\circ)$, and during the RCM control period, the needle posture can be adjusted from 60° to 30° (θ_{2r}, after the RCM period is finished). The needle can stop at any time during this period if the doctor thinks that the needle has a good posture for insertion of the guide wire. Each DOF is set with a displacement function based on the values of Δd_1, Δd_2, and Δd_3, as described in Section 3.3. The simulator can then be started to record the test results.

(2) *Experiments using the real manipulator system*: The experimental environment in ADAMS is idealized, i.e. there are no mechanical errors or control errors. However, mechanical and control errors are unavoidable in the real world. Therefore, we need to verify the accuracy of the proposed method using a real puncture manipulator.

Experimental environment: The experimental environment is as shown in Figure 7. The puncture needle model used in this experiment is the 21G (0.80 mm × 76 mm). Because we do not need to consider the deformation of the tissues or the needle in the vessel model, water was used to replace the human body model.*Parameter setting:* An image of the needle in water can be acquired using the ultrasound probe. The average depth of the subclavian vein is approximately 20 mm. To ensure that the experimental environment is as close as possible to the actual surgical environment while also making the parameters universal, the puncture angle was set at 45° and the puncture target was M_R $(-22, -8)$ (units: mm). The needle angle after RCM was set at 30°. To provide a better indication of the target, a virtual blood vessel was added, and *P* is on the centre line of this virtual blood vessel.

Posture calculation of needle in {***B***}. Because the positional relationship between the manipulator and the ultrasound probe is known, each pixel in the ultrasound image has corresponding coordinates in {***B***}. Based on the real sizes of the ultrasound probe and the ultrasound image, we obtained a square with dimensions of 10 mm × 10 mm containing 270 × 270 pixels. Therefore, one pixel has dimensions of 0.037 mm × 0.037 mm, which means that the needle tip position tracking accuracy is 0.037 mm. By tracking the pixel coordinates of the needle in the ultrasound image, we can track the needle tip trajectory and obtain the needle posture information.

Based on the settings above, the manipulator system was controlled to finish the puncturing and RCM movement. Table 2 shows the parameters used and how to calculate them.

After all parameters were set correctly, the bottom of the ultrasound probe was put into the water, the robot was fixed using a support frame, and the system was switched on. The experiment was repeated four times.

Table 2. Main parameters used in the experiments.

Parameter type	Parameters	Origins of parameters (from experiments or calculations)
Control parameters	$\Delta d_1, \Delta d_2, \Delta d_3$	Displacements of three motors, calculated from Equations (10–21).
Target parameters	θ_{2p}	Puncture angle. Set by doctor.
	θ_{2r}	Needle angle after RCM. Set by doctor.
	$M_S(-V,-H)$	Puncture target location in simulation experiment. Set by doctor, based on the simulation.
	$M_R(-V,-H)$	Puncture target location in real robot experiment. Selected from ultrasound image by doctor.

4.2. Simulation in ADAMS

Figure 8 shows the motion series of the robot in the simulation and Figure 9 indicates the needle tip coordinates in $\{\boldsymbol{B}\}$, the movement distances of the three joints and the angle θ_2 during the experiments. The initialized state of the robot is shown in Figure 8(1), where the needle is oriented perpendicular to the bottom plane of the ultrasound probe (horizontal plane) in this state. From Figure 8(1) to 8(4), the needle's posture was adjusted to set the manipulator into the preset puncture state (puncture angle: 60°). This process can also be seen in period T1 in Figure 9. During this period, the rotation joint (joint 2) of the manipulator will be pushed by motor 2 until the rotation angle reaches 60°. Figure 8(5) to 8(7) indicate that the manipulator is implementing the puncture action. During this period, the needle will pierce the skin and the blood vessel before reaching the puncture target. The period T2 in Figure 9 shows the movements of the three joints during the puncture process. Figure 9(a)

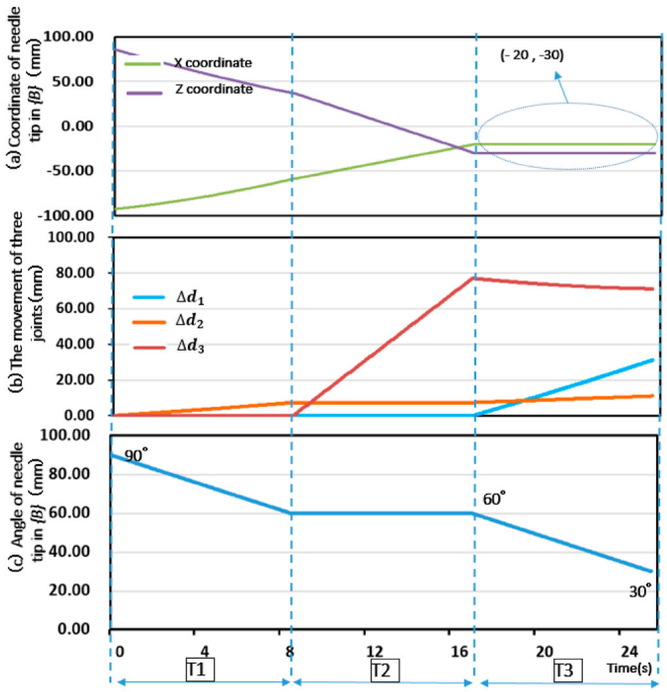


Figure 9. Parameters used in simulator.

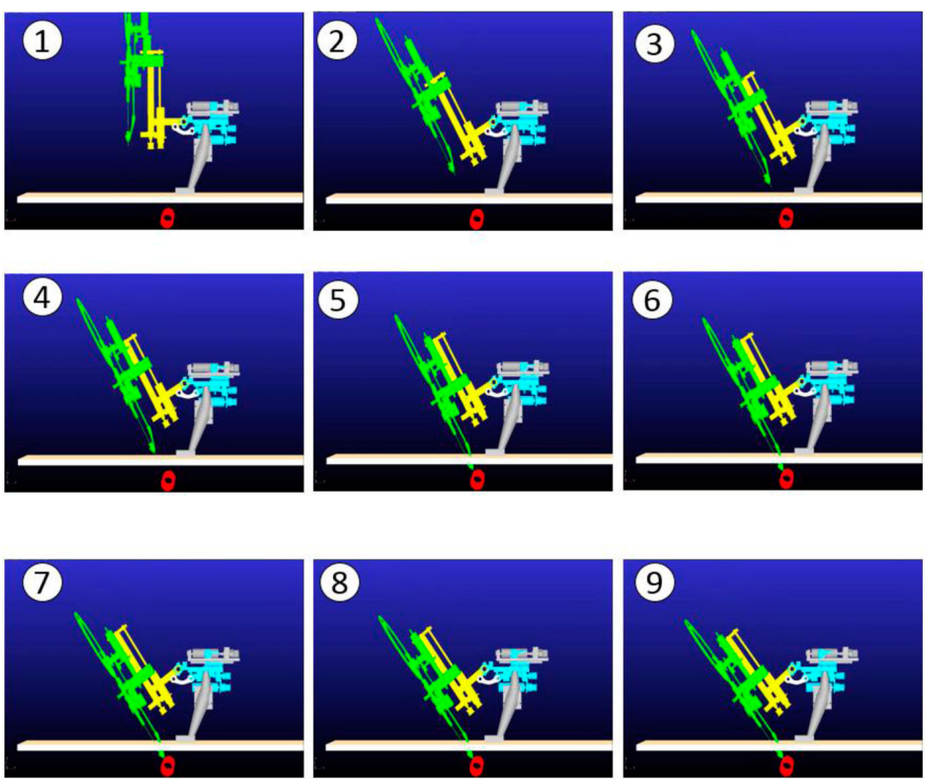


Figure 8. Puncture and RCM in ADAMS.

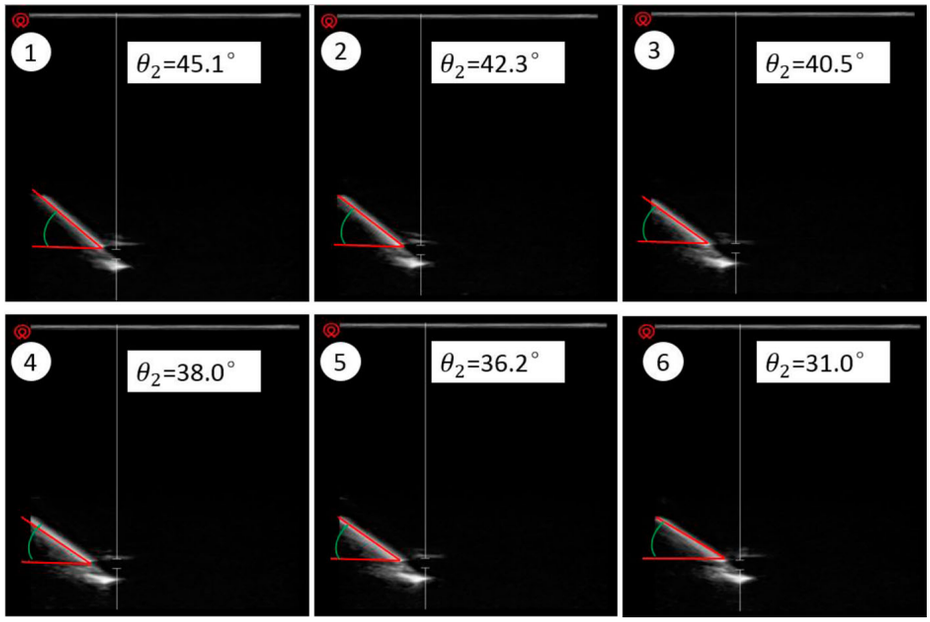


Figure 10. Angle changes of the needle during RCM control.

shows that the needle tip arrived at the puncture target point $M_5(-20, -30)$ at the end of period T2. After the puncture movement, RCM control was performed and the needle posture was adjusted without changing the needle tip position, as shown in Figure 8(8) and Figure 8(9). RCM control of the manipulator can also be observed in Figure 9, as shown in period T3. It was found that the three joints move simultaneously to adjust the needle posture precisely while keeping the needle tip in the same place.

Figure 9(c) shows the angle change in the needle during the tests. This simulation shows that the change in the needle angle is fully in line with the theoretical value from the experiments. The initial manipulator state ($\theta_2 = 90^\circ$), the preset puncture angle ($\theta_2 = 60^\circ$) and the target angle ($\theta_2 = 30^\circ$) can all be found in this figure.

The results of simulation of the puncture operation and the RCM operation in ADAMS agree completely with the theoretical results, thus verifying the correctness of our algorithms and the proposal of RCM control.

4.3. Experiment using real robot

If the needle is shown in an ultrasound image, the needle angle can be acquired by image processing. It is difficult to obtain the real angle between the needle and the blood vessel accurately if the puncture needle cannot be scanned using an ultrasound probe. Therefore, we simply focused on the RCM period (T3 in Figure 9) in the experiments using the real manipulator system.

Based on the experimental settings, the puncture angle is 45° and the RCM target angle is 30°. The angular

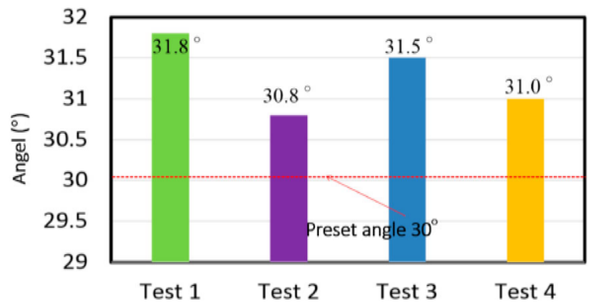


Figure 11. Final needle angles after RCM control in four tests.

change in the needle in one test during RCM control is shown in Figure 10(1–6). The figure shows that the needle angle at the beginning and the target angle are almost same as the theoretical values (45.0° and 30.0°, respectively). The final needle angles after RCM control in four tests are shown in Figure 11. When compared with the theoretical values, the errors are 1.8°, 0.8°, 1.5° and 1.0°.

The needle point positional errors during RCM control are shown in Figure 12. It is clear that the positional errors during RCM control do not exceed 0.8 mm in any of the tests. Theoretically, the needle point should remain at the puncture target during RCM operation and we infer that these experimental errors were caused by mechanical and control errors of the manipulator system. Figure 3 shows that the distance from the needle tip to the rotation joint is long. This type of structure acts as an error amplifier. Therefore, even if only a small mechanical or control error exists in this system, the final needle angle and position will differ significantly from the preset values.

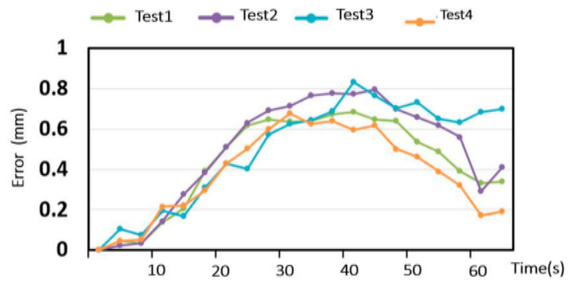


Figure 12. Positional error of needle tip during RCM control.

It was also found that the needle tip positional error tended to increase initially and then decrease during these tests. We believe that this behaviour is caused by mechanical and control errors. The mechanical error means that the real RCM point is not exactly at the needle tip and thus the needle tip will rotate around the real RCM point. However, because of the control accuracy and the error amplifier structure, the position of the real RCM point also changes slightly during the tests. The final needle tip trajectory is thus similar to a small ellipse and it appears as this error tendency.

Although some errors exist in these experiments, when compared with the diameter of the central vein and the accuracy of manual puncture surgery, the angular and positional errors in the RCM operation are both acceptable in real puncture surgeries. The results of both the ADAMS simulations and the real puncture manipulator tests indicate that the RCM control method can meet the design requirements. RCM can provide higher angular and positional control of puncture needles during surgery and has good application prospects.

5. Conclusions

In this paper, we introduced a puncture control method to improve the efficiency and safety of robot-assisted CVC surgery. To prevent puncture needles from piercing or detaching blood vessels, we proposed and designed an operation for angular and positional adjustment of needles called RCM. During RCM processing, the needle tip will be held at a fixed position while the posture is adjusted until it reaches the required state for guide wire insertion. Experiments were conducted in the ADAMS simulator and a real puncture manipulator system. The results indicated that the simulation results agreed completely with the theoretical values. Angular and positional errors existed in the real puncture manipulator experiments, but when compared with the size of the central vein and the accuracy of manual puncture surgeries, these errors were acceptable. Two experiments verified the feasibility of the RCM control method.

As discussed earlier, the deformation of the tissues and the needles are not considered in this paper; however, this deformation is important and will affect puncture and needle adjustment accuracies during real surgeries. The algorithms and settings used in this paper cannot be applied directly to a surgical procedure. Therefore, a deformation model will be added and new experiments will then be implemented in future work.

Disclosure statement

No potential conflict of interest was reported by the authors.

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